



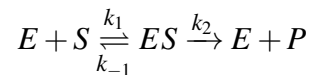
Division of Biological Sciences & Quantitative Ecology

Problem Set 3: Enzyme Kinetics Derivations and Predator-Prey Dynamics

Due: Friday, October 31 at 17:00 • Total Score: 100 Points

Problem 1: Quasi-Steady-State Derivation of Michaelis-Menten Kinetics (50 Points)

Consider the standard single-substrate enzymatic reaction scheme:



where $[E]_0 = [E] + [ES]$ denotes total enzyme concentration.

- Apply the Briggs-Haldane Quasi-Steady-State Assumption (QSSA), $\frac{d[ES]}{dt} = 0$, to express $[ES]$ in terms of $[E]$ and $[S]$.
- Define the Michaelis constant $K_m = \frac{k_{-1} + k_2}{k_1}$ and maximum velocity $V_{\max} = k_2[E]_0$. Derive the velocity rate equation $v = \frac{V_{\max}[S]}{K_m + [S]}$.

Detailed Solution & Mathematical Derivation

Part (a): Rate of change of the intermediate complex $[ES]$ is:

$$\frac{d[ES]}{dt} = k_1[E][S] - (k_{-1} + k_2)[ES] = 0 \implies k_1([E]_0 - [ES])[S] = (k_{-1} + k_2)[ES]$$

Part (b): Rearranging for $[ES]$:

$$[ES] \left(\frac{k_{-1} + k_2}{k_1} + [S] \right) = [E]_0[S] \implies [ES] = \frac{[E]_0[S]}{K_m + [S]}$$

The reaction velocity is the rate of product formation $v = \frac{d[P]}{dt} = k_2[ES]$:

$$v = k_2 \frac{[E]_0[S]}{K_m + [S]} = \frac{V_{\max}[S]}{K_m + [S]} \quad \blacksquare$$

Problem 2: Equilibria of the Lotka-Volterra Predator-Prey System (50 Points)

The non-dimensionalized Lotka-Volterra predator-prey system is given by:

$$\frac{dN}{dt} = rN - aNP, \quad \frac{dP}{dt} = baNP - mP$$

- Find all fixed points (N^*, P^*) where $\frac{dN}{dt} = 0$ and $\frac{dP}{dt} = 0$.
- Linearize the system around the non-trivial co-existence equilibrium and determine the eigenvalues of the Jacobian matrix.

Detailed Solution & Mathematical Derivation

Part (a): Setting derivatives to zero:

$$\begin{aligned} N(r - aP) &= 0 \implies N = 0 \text{ or } P^* = \frac{r}{a} \\ P(baN - m) &= 0 \implies P = 0 \text{ or } N^* = \frac{m}{ba} \end{aligned}$$

Fixed points are $(0, 0)$ (extinction) and $(N^*, P^*) = (\frac{m}{ba}, \frac{r}{a})$ (co-existence).

Part (b): The Jacobian matrix is:

$$\mathbf{J}(N, P) = \begin{pmatrix} r - aP & -aN \\ baP & baN - m \end{pmatrix}$$

Evaluated at (N^*, P^*) :

$$\mathbf{J}(N^*, P^*) = \begin{pmatrix} 0 & -\frac{am}{ba} \\ ba\frac{r}{a} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{m}{b} \\ br & 0 \end{pmatrix}$$

The characteristic equation is $\det(\mathbf{J} - \lambda \mathbf{I}) = \lambda^2 + mr = 0 \implies \lambda = \pm i\sqrt{mr}$. Since eigenvalues are purely imaginary, the non-trivial fixed point is a neutrally stable center surrounded by periodic closed orbits. ■